

Revision: Different Photosynthetic Responses to Heat and Light Favour Green and Red over Brown Macroalgae in the Mediterranean Sea

Summary: We would like to thank the editor and the two reviewers for their constructive comments. In the following, we are addressing all comments in a point-to-point response in order to improve the manuscript. In addition to numerous textual clarifications, added references, and refinements of style and structure suggested by the reviewers, we made several major revisions in line with the editor’s requests. Specifically, we clarified the taxonomic uncertainty surrounding *Cystoseira* and the limitations this imposes on the generalizability of our findings, while justifying the genus-level approach with supporting evidence. The Discussion was expanded to interpret the significant three-way interaction (Temperature × Light × Taxon), highlighting divergent taxon-specific trajectories. To avoid overstatement, we revised language on resilience, framing interpretations of the P:R model as observed trends. Finally, the experimental design was clarified by explaining how incubation chambers were randomized across water baths to ensure independence of replicates. Collectively, these revisions improve clarity, methodological transparency, and the overall robustness of the manuscript.

With best regards,

Leonie Hesse on behalf of all co-authors

RESPONSES TO COMMENTS MADE BY THE EDITOR

Reviewer	Manuscript Line	Comment	Response
Editor	/	clarify the taxonomic uncertainty regarding <i>Cystoseira</i> and the limitations this imposes on the generalizability of results	We agree that the taxonomic uncertainty of <i>Cystoseira</i> is an important limitation for generalizability. As noted in our response to Reviewer 1, we have added a paragraph to the Discussion (lines 473-485) acknowledging this constraint while explaining why the genus-level approach remains appropriate.
Editor	/	clarify the interpretation of taxon-specific responses under dual stress, especially given the significant three-way interaction, which is not discussed	We now discuss the significant three-way interaction between temperature, light, and taxon to support the interpretation of taxon-specific trajectories (Discussion, lines 420-444; see also our reply to Reviewer 1).
Editor	/	clarify the use of strong language around resilience without formal inference	We concur that strong language on resilience should be limited to formal inferences. Accordingly, as noted in our response to Reviewer 1, we have revised all statements based on the P:R model to emphasize observed trends rather than overstate conclusions, with clarifications in Results (lines 314, 318, 322) and Discussion (lines 370–372, 454-456).
Editor	/	clarify the experimental independence of replicates, as it is not fully clear whether incubation chambers were randomized across independent temperature/light baths	We agree that clarifying the independence of replicates is important. As also noted in our response to Reviewer 1, we used independent baths and randomized the placement of incubation chambers across them. This clarification has been added to the Materials & Methods section (line 155).

RESPONSES TO COMMENTS MADE BY REVIEWER 1

Reviewer 1	/	The decision to analyze <i>Cystoseira</i> only at the genus level is understandable due to morphological ambiguity. However, as responses can vary considerably between species within this genus, a brief justification and acknowledgment of the limitations this imposes on the generalizability of your results should be added (e.g., under “Discussion section”).	We thank the reviewer for raising this point. We have added a paragraph to the Discussion acknowledging that analyzing <i>Cystoseira</i> at the genus level limits transferability to individual species, while also explaining why this approach is appropriate and supported by statistical and descriptive evidence of limited within-genus variability. Please refer to Discussion, lines 473-485 of the revised manuscript. We also additionally explored within-group variability descriptively by calculating dispersion metrics (standard deviation, MAD, coefficient of variation) and comparing variances of <i>Cystoseira</i> against the other taxa across temperature–light treatments. <i>Cystoseira</i> exhibited variance comparable to, and in most conditions lower than, <i>Phyllophora</i> and <i>Flabellia</i>, with only isolated cases of higher variance. We added a sentence stating this in the methods (see lines 127-130). Together with our candidate mixed effect model showing negligible random-effect variance for the taxon term, we thus believe that our within-taxon variability in <i>Cystoseira</i> samples was limited and thus inferences to be meaningful.
Reviewer 1	/	You appropriately refrained from reporting formal statistical inference for the P:R model due to mild assumption violations. However, some conclusions (e.g., “<i>Flabellia</i> was the most resilient taxon”) may overstate certainty, especially since they rely partly on descriptive patterns in P:R ratios. I recommend clarifying that this interpretation is based on observed trends and supported by significant differences in net photosynthesis.	We have revised all statements derived from the P:R model to avoid overstatement, clarifying that these interpretations are based on observed trends in P:R ratios and supported, where applicable, by statistically significant differences in net photosynthesis. Please refer to Results, lines 314, 318, 322 and Discussion, lines 370–372, 454-456.
Reviewer 1	/	The manuscript is generally well-written, but there are minor grammatical and syntactical issues throughout. Careful proofreading is recommended (see specific examples below).	The manuscript has been carefully proofread, and all grammatical and syntactical issues have been corrected, including the specific examples noted by the reviewer.

Reviewer 1	/	In the Results section, several numerical values (particularly for net photosynthesis and respiration) are reported without their corresponding units (e.g., $\text{nmol O}_2 \text{ m}^{-2} \text{ s}^{-1}$). For instance, values such as “net photosynthesis increased to 21.44” or “respiration remained elevated (-10.3 ± 0.85)” appear without specifying units, despite being central to the interpretation of physiological responses. While these units are defined in the Methods and figures, I recommend including them.	We agree that consistency in displaying units will improve readability. We added units at all presented numerical values in the result section (for respiration and net photosynthesis). We did not explicitly repeat the unit "P:R" when presenting P:R ratios, as the term "P:R" was always mentioned before; e.g., "P:R was 3.64 ± 0.34" (line 287).
Reviewer 1	/	Phyllophora showed a significant increase in net photosynthesis under high temperature and light), this positive response to dual stress warrants discussion, as it contradicts general expectations of decline. Conversely, while Cystoseira performed poorly under high light, its performance under moderate conditions was not the weakest and could also be briefly acknowledged. Furthermore, the significant three-way interaction (Temperature $\times$ Light $\times$ Taxon; $p = 0.023$) is not mentioned in the Discussion but should be, as it supports the interpretation of taxon-specific trajectories.	We have now added content to the Discussion to interpret the observed taxon-specific responses of brown and red algae. The revision also incorporates the reviewer's points regarding the response of Phyllophora under dual stress, the performance of Cystoseira under moderate conditions, and the significance of the three-way interaction (see Discussion, lines 420-444).
Reviewer 1	91	such as <i>Ulva</i> spp. [26] or <i>Sargassum muticum</i> displace native communities..." → Consider rephrasing for grammatical consistency: "...such as <i>Ulva</i> spp. or <i>Sargassum muticum</i>, which displace native communities..."	We thank the reviewer for this suggestion. The sentence has been rephrased for grammatical consistency in the revised manuscript (see Introduction, lines 91-92).
Reviewer 1	/	Please clarify whether incubation chambers were placed in separate, independently controlled water baths per replicate, or if all six chambers for a given treatment were incubated together in a single shared bath. Since each chamber is treated as an independent replicate ($n = 6$ per treatment), this distinction is critical to rule out pseudoreplication and ensure the statistical independence.	We agree that clarifying the incubation setup is essential to address concerns about pseudoreplication. While individual water baths for each incubation chamber were not available, we used four independent water baths and randomly distributed the chambers within them. This clarification has now been added in the Materials & Methods section (line 155).

Reviewer 1	225–236	In the section reporting the three-way interaction between temperature, light, and taxon (Results, lines ~960–975), only <i>Phyllophora</i> and <i>Cystoseira</i> are discussed explicitly. Although the results for <i>Flabellia</i> are presented later in the section, it would improve clarity and consistency to summarize <i>Flabellia</i>'s performance under dual stress in the same paragraph where the interaction is first introduced. This would reinforce the full interpretation of the interaction term and ensure all taxa are treated equally in the main effects analysis. Additionally, the interpretation that temperature and light effects are “strongly non-additive” may require refinement. While <i>Cystoseira</i> clearly exhibits a non-additive negative response under dual stress, <i>Phyllophora</i> appears to benefit from the combination of high light and high temperature (i.e., additive or even synergistic enhancement). <i>Flabellia</i> also maintains high performance under dual stress. Therefore, I suggest rephrasing the conclusion of this paragraph to clarify that the interaction between stressors is taxon-specific, rather than uniformly non-additive. From line 220: Within the Results section, I noted multiple instances where specific values (e.g., net photosynthesis under control or dual stress conditions, respiration, or P:R ratios) are repeated across adjacent paragraphs. For example, baseline photosynthetic rates are mentioned twice for <i>Cystoseira</i> and <i>Phyllophora</i>, and the decline in <i>Cystoseira</i> under high light is presented with slightly different values. To improve clarity and reduce redundancy, I recommend streamlining these descriptions by reporting each result once, then referring back to them when needed (e.g., “as previously described”). This will help tighten the narrative and avoid potential confusion due to slightly varying numerical estimates. Example: Line 231-232: <i>Cystoseira</i> showed a marked decline in net photosynthesis under high light alone (-12.2 ± 2.3 nmol...) //Line 257-258: High light alone at control temperature (21 °C, 760 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$) caused a	We have revised the Results section to integrate <i>Flabellia</i>'s performance under dual stress directly into the paragraph introducing the three-way interaction, ensuring all three taxa are discussed consistently at the point where the interaction is first described (lines 252-257). We also refined our conclusion in this paragraph to emphasize that the interaction is taxon-specific: negative and non-additive in <i>Cystoseira</i>, additive/synergistic in <i>Phyllophora</i>, and buffering in <i>Flabellia</i> (see Results, lines 255-257). Finally, to improve clarity and avoid redundancy, we streamlined the Results section by removing repeated numerical values and instead referring back to previously reported results (e.g., “as previously described”). We agree that these suggested edits improve clarity and readability.
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		significant drop in photosynthesis to 7.75 ± 1.67 ... Other example: Line 251: ... statistically indistinguishable from <i>Phyllophora</i> (19.72 ± 1.67 , $p = 0.994$...//Line 267: At baseline (control temperature, control light), photosynthesis was 19.72 ± 1.67 (Fig. 2)	
Reviewer 1	358	the sentence “As a red alga, <i>Phyllophora</i> possesses chlorophyll a and phycobiliproteins arranged in phycobilisomes” may be misleading. Chlorophyll a is present in all photosynthetic algae, including green and brown taxa. Consider rephrasing to emphasize that what distinguishes red algae is the presence of phycobiliproteins (e.g., phycoerythrin and phycocyanin) organized in phycobilisomes, which optimize light harvesting in low-light environments.	We thank the reviewer for the clarification. The sentence now specifies that chlorophyll a is common to all photosynthetic algae, and highlights phycobiliproteins in phycobilisomes as the distinguishing feature of red algae (see Discussion, lines 394-397).
Reviewer 1	22	“...using specimen collected...” → “specimens collected”	The phrase has been corrected to “specimens collected” in the revised manuscript (see Abstract, line 22).
Reviewer 1	186	“...nested model comparisons using likelihood ratio tests (ANOVA)...” → clarify that you used likelihood ratio tests via ANOVA comparisons.	The sentence has been revised to clarify that we used likelihood ratio tests implemented via ANOVA model comparisons (see Material & Methods, lines 200-202).
Reviewer 1	410	epibionts ^{45,46} . Please add a space and use parentheses to improve clarity and ensure proper formatting, e.g., “epibionts [45,46]”.	The citation has been corrected in the revised manuscript (see Discussion, line 462).
Reviewer 1	415	nutrient limitation ⁵ . Please add a space and use parentheses to improve clarity and ensure proper formatting, e.g., “limitation [5?]”.	The citation has been corrected in the revised manuscript (see Discussion, line 467).

RESPONSES TO COMMENTS MADE BY REVIEWER 2

Reviewer 2	/	Is there any evidence that the light levels used in this study, especially the ‘high light’ conditions are expected to cause stress to the seaweeds? Especially the red and green which to me did not seem to be negatively affected by the high light (figure 2). What are the ambient light levels in the sampling site throughout the year? To properly determine the photosynthetic limits of a species a Photosynthesis/Illumination curve is required.	We agree that contextualizing the light levels used in our treatments is important. According to the literature, photoinhibition has been reported in several Mediterranean red and green macroalgae already at moderate irradiances, whereas more surface-adapted taxa can tolerate substantially higher intensities (Häder & Figueroa, 1997). We acknowledge that full P–E curves would be necessary to rigorously determine the photosynthetic limits of each species, and we identify this as an important direction for future research. Unfortunately we have no <i>in-situ</i> filed data on ambient light conditions, but to approximate ambient light conditions at the study site, we retrieved remotely sensed daily surface photosynthetically active radiation (PAR) data from NASA’s POWER reanalysis product (CERES SYN1deg) and converted it to average daytime irradiance ($\mu\text{mol photons m}^{-2} \text{s}^{-1}$). The surface PAR at the study site during the experimental period (September 2021) fluctuated between ~200–270 $\mu\text{mol m}^{-2} \text{s}^{-1}$, aligning well with our control and medium-light treatments. These values give incident radiation at the surface and actual benthic light availability is expected to be lower. Therefore, we believe our control treatment likely reflects a conservative estimate of natural <i>in situ</i> conditions, while the high-light treatment represents a realistic upper-limit stress scenario.
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Reviewer 2	/	Oxygen levels were measured in air-tight metabolic chambers after exposure to the experimental conditions for two hours. This measurement may be problematic. The authors calculated the oxygen evolution rates as if they were linear throughout the two hours. This is not necessarily true, e.g – the seaweeds may have suffered from photoinhibition and decreased their photosynthetic activity after a shorter time period. The CO₂ in the metabolic chamber may have been spent, thus limiting photosynthesis (there was no report of pH values at the start and end of the incubation. In natural conditions there is a constant exchange of seawater around the seaweed. In addition, CO₂ solubility decreases with rising temperatures (the authors also mention this in the discussion), so during the experimental setup it is necessary to bring the water to the desired temperature before incubation the samples in the light. It is not clear from the methods if this was done. Also, the O₂ evolution rates were normalized to surface area, this will probably work well for species like <i>Flabiella</i> but for a bulky seaweed like <i>Cystoseira</i> this may be less accurate. Perhaps it would have been better to normalize to weight (especially dry weight) or to chlorophyll levels, which is a common practice. Unfortunately, I didn't see and report on pigment levels of the species used in the current experiment. I hope that the authors saved the samples in a dry or frozen state. I would then recommend them to extract the photosynthetic pigments, show their abundance throughout the experiments, and also normalize the O₂ evolution rates the Chl levels.	We thank the reviewer for these points. We agree that O₂ rates may not always be linear and can decrease due to photoinhibition. Assuming linear rates and standardizing over time provides a conservative estimate, avoiding overestimation from short-term peaks, while reflecting mean performance and enabling comparability across taxa. This approach is widely applied in algal research (Häder & Figueroa, 1997), though future studies could incorporate non-linear dynamics more explicitly. In addition, the duration of our incubations was kept as short as possible in order to avoid any excess consumption or production of carbon dioxide. Oxygen values were not below 50 % saturation after dark incubation and not above 120 % saturation in the light in order to avoid pronounced hypoxic or hyperoxic conditions that may have influenced the physiology of the algae. This information is now been given in the revised manuscript in lines 162 to 164. Secondly, we confirm that target water temperatures were reached prior to the start of the experiment at the respective light levels, before the chambers were closed (see Material & Methods, line 156). Thirdly, we have added an explanation of the surface area quantification method to clarify its precision. Specimens were flattened on a sheet of paper, photographed, and the surface area was quantified through automated substrate selection using dedicated software (see Material & Methods, lines 170–173). Standardizing oxygen production to surface area is a well-established practice, comparable to the reviewer's suggested approaches (DW or chlorophyll levels). Since no frozen or dried specimens were retained, subsequent analyses were not possible. Nevertheless, including complementary metrics such as biomass or pigment content would indeed be valuable to explore in future research.
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Reviewer 2	/	In the discussion they say that this was already known (line 326). Also the authors suggest several possible mechanisms to explain the response of the different seaweeds to the light/temperature regimes, relying on available literature. Instead of relying on literature, some research efforts could have been made to make validation tests in the current experiment (e.g – measuring pH or DIC levels in the metabolic chambers to support the claim in line 341, or a simple pigment extraction to support the claim in line 354. Chlorophyll fluorescence measurements would have also been helpful to determine important values such as NPQ (line 330) but such equipment is not always available). To conclude, the results may be interesting, especially regarding <i>Cystoseira</i> , but I think more efforts are required to provide better results and more insight into the response of the seaweeds to the light/temperature regime.	We thank the reviewer for these valuable suggestions. We agree that additional measurements such as pH or DIC in the metabolic chambers, pigment extractions, or chlorophyll fluorescence would have further strengthened our interpretations. Unfortunately, such analyses were not feasible within the scope of the current study due to logistical and equipment constraints, and no frozen or dried specimens were retained for subsequent analyses. We therefore relied on well-established methods and complementary literature to interpret the observed responses. We acknowledge these as limitations and highlight them as promising directions for future research.
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